

A top-down view of a workspace. In the top left, a silver laptop is partially visible. To its right is a white computer mouse. Above the mouse is a cup of dark coffee. In the bottom left, there is a dark brown leather notebook with a white pen resting on it, and a pair of tortoiseshell glasses. The background is a light teal color with three overlapping teal squares of different shades.

AIClub





Data Exploration & Preprocessing

What is Data Exploration?

After collecting data, we must:

- Understand the data
- Analyze the data
- Prepare the data for modelling
- This stage is called Data Exploration / Preprocessing
- Raw data cannot be directly used for training



Why is Data Exploration Important?

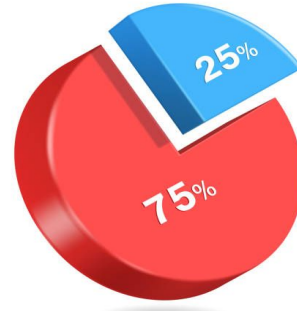
- Identifies errors in data
- Finds patterns and trends
- Improves data quality
- Helps build accurate AI models
- Clean and prepared data = Better model performance



Steps in Data Exploration

Data Exploration includes:

1. Data Cleaning
2. Data Splitting
3. Pattern Identification
4. Data Visualization



Step 1 - Data Cleaning

- Removing incorrect data
- Fixing missing values
- Removing duplicate entries
- Correcting spelling or formatting errors
- Example: If attendance is written as 105%, it must be corrected.



Why Data Cleaning is Needed?

If data is:

- Incomplete
- Incorrect
- Inconsistent

Then AI predictions will be wrong.

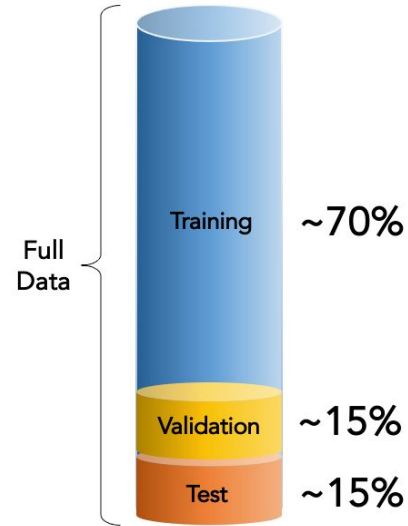
“Garbage In, Garbage Out”



Step 2 - Data Splitting

After cleaning, data is divided into parts:

1. Training Data
2. Testing Data
3. Validation Data



Training, Testing & Validation

Training Data:

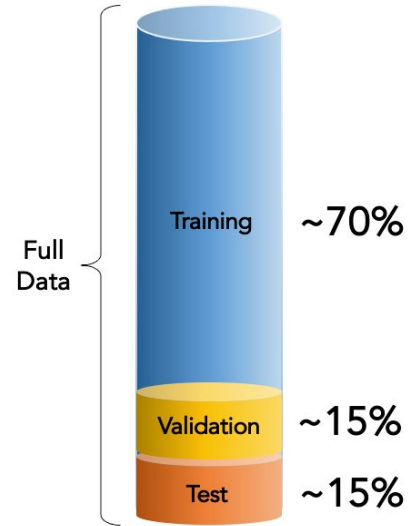
- Used to teach the model
- Largest portion (usually 70–80%)

Validation Data:

- Used to tune the model

Testing Data:

- Used to check final accuracy
- Prevents overfitting and ensures reliability.



Step 3 - Pattern Identification

Pattern Identification means:

- Finding relationships in data
- Observing trends
- Identifying similarities
- Example:
 - More study hours → Higher marks
 - Higher temperature → More ice cream sales
- Patterns help AI learn.



Example of Pattern Identification

Problem: Student Performance

Observation:

- Students with $> 90\%$ attendance score higher.

Attendance and marks are related. This is a pattern.



Step 4 - Data Visualization

- Representing data using charts and graphs
- To make patterns easier to understand.
- Common Types:
 1. Bar Graph
 2. Pie Chart
 3. Line Graph



Example – Visualizing Student Data

If we collect:

- Study hours
- Marks
- Attendance

We can:

- Plot bar graphs for comparison
- Use line graphs for trends
- Use pie charts for grade distribution



How to Choose the Correct Graph?

Choosing the right graph helps us understand data clearly.

Ask these questions:

1. What type of data do I have?
2. Am I comparing categories?
3. Am I showing trends over time?
4. Am I showing proportions or percentages?



The purpose decides the graph.

When to Use a Bar Graph?

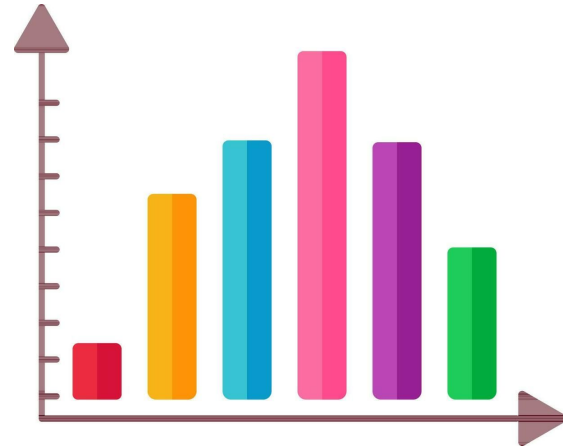
Bar Graph

Use when:

- Comparing different categories
- Showing differences between groups

Examples:

- Marks in different subjects
- Number of students in each section
- Sales of different products
- Best for comparing values.



When to Use a Line Graph?

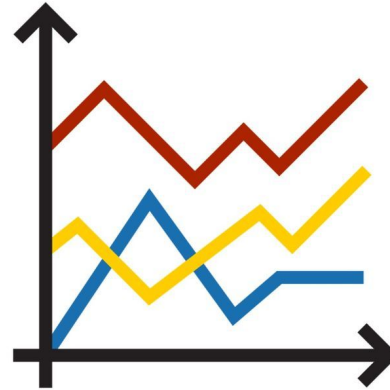
Line Graph

Use when:

- Showing trends over time
- Observing increase or decrease

Examples:

- Temperature across days
- Attendance across months
- Stock prices over time
- Best for showing change and trends.



When to Use a Pie Chart?

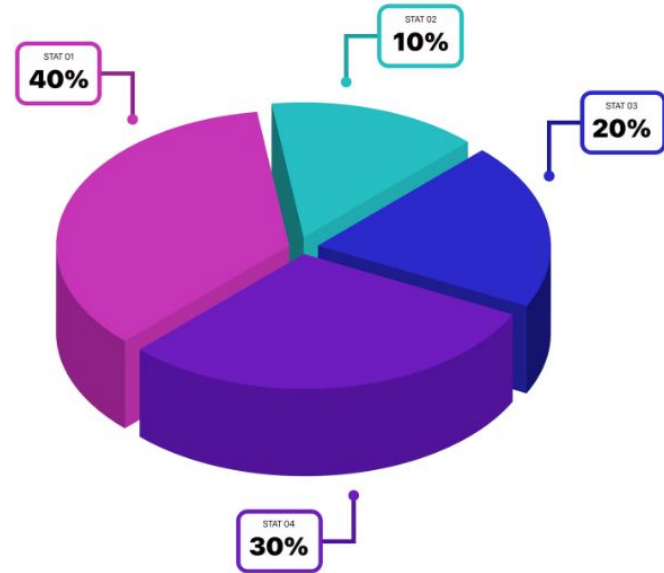
Pie Chart

Use when:

- Showing percentages
- Displaying parts of a whole

Examples:

- Grade distribution
- Budget allocation
- Market share
- Best for proportion comparison.



Quick Graph Selection Guide

Situation	Best graph
Compare categories	Bar graph
Show trends over time	Line graph
Show percentage distribution	Pie graph

- **Rule of Thumb:**
 - Comparison → Bar
 - Trend → Line
 - Proportion → Pie

How Data Exploration Helps AI

Data exploration:

- Improves data quality
- Reduces errors
- Identifies useful features
- Makes modelling more accurate
- Proper preprocessing leads to better AI performance.



Key Takeaways

- ✓ Data Exploration prepares data for modelling.
- ✓ Cleaning removes errors.
- ✓ Data is split into training, validation, and testing sets.
- ✓ Patterns help AI learn relationships.
- ✓ Visualization makes data easier to understand.

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THANK YOU



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